

Rigidity of the Standard-Model Charge Lattice

A Fixed-Scope Reconstruction Theorem for Charge Quantization

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Abstract

Charge assignments in the Standard Model are usually fixed by representation choices, hypercharge assignments, and anomaly-cancellation constraints. This paper proves a fixed-scope reconstruction theorem about when such assignments can function as standing-preserving bookkeeping data. For a declared Standard-Model chiral skeleton with charged Yukawa operators, admissible charge labels are required to preserve interaction neutrality, conjugation, closed redescription transport, and the anomaly/tensor witness ledger. Under these conditions, every standing-bearing charge realization factors through a discrete additive charge lattice that is unique up to admissible redescription, global normalization, and overall sign.

The physics-facing realization is the one-generation electroweak hypercharge system. Yukawa compatibility and anomaly constraints collapse the hypercharge assignments to a one-parameter family. The remaining parameter is absorbed into the $U(1)_Y$ coupling; after fixing a primitive unit, the electric charges occupy the familiar lattice, conventionally $\frac{1}{3}\mathbb{Z}$ when $Q(e) = -1$. Fractional-looking Standard-Model quark charges are therefore admissible lattice points, not counterexamples. What is excluded is non-lattice fractional interpolation or arbitrary same-scope charge deformation.

The result does not derive the gauge group, numerical gauge couplings, or all ultraviolet completions from no premises, and it does not deny larger-domain beyond-Standard-Model theories. It establishes equivalence-class uniqueness inside a fixed admissible Standard-Model reconstruction category. A charge-transport certificate, universal charge quotient, partial-operator obstruction, and rival-exhaustion argument show that any proposed same-scope alternative must factor through the same charge quotient, collapse to bookkeeping, fail admissibility, or change scope.

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1 Introduction and theorem target

The target question is not whether charged particles exist, whether gauge theory is useful, whether anomaly cancellation is standard, or whether the observed Standard Model has quark charges $2/3$ and $-1/3$ in electron units. The target question is prior and narrower:

Within a fixed admissible Standard-Model bookkeeping skeleton, can charge labels be continuously deformed, fractionally interpolated, shifted, or replaced by inequivalent same-scope lattices while preserving standing-bearing transport and all declared charge witnesses?

The answer proved here is no. A standing-bearing charge label must be stable under admissible redescription, closed under charge composition and conjugation, compatible with the declared interaction witnesses, and invariant under closed redescription transport. Those requirements force a discrete additive charge lattice. In the Standard-Model chiral skeleton with one Higgs doublet and the declared charged Yukawa operators, the hypercharge equations collapse to the standard one-parameter family. The parameter is not a physical continuum of relative charge assignments; it is normalization of the abelian generator, absorbed into g_1 until a primitive-unit convention is fixed.

The present formulation distinguishes three issues that are often conflated: rational coordinates inside a fixed primitive lattice, non-lattice interpolation between lattice points, and normalization/sign convention for the abelian generator. Fractional-looking coordinates are admissible when they are lattice points after primitive-unit normalization. Additive offsets are not normalization; they change neutrality and tensor compatibility. The charge lattice is therefore treated as an additive normal form rather than as a loose discrete label set.

Core claim. In a fixed admissible charge scope, every standing-bearing charge assignment factors through a unique discrete additive lattice up to admissible redescription, primitive-unit normalization, and global sign. A rival that changes relative charge positions changes tensor witnesses or scope; a rival that changes no witness is skin; a rival that selects hidden labels, interpolates non-lattice fractions, repairs failed transport, or adds an unanchored lattice is inadmissible.

Main sequent. On the declared charge domain,

$$\begin{aligned} &\text{Adm}(\mathcal{S}_{\text{chg}}), \quad \text{Stand}(q), \quad \text{Transp}_{\text{cl}}(q), \quad \text{Wit}_{\text{chg}}(q) \\ &\implies q \text{ factors through } \Lambda_{\text{chg}} \cong u\mathbb{Z} / \sim_{\text{norm,sign}} . \end{aligned}$$

For the Standard-Model electroweak skeleton with $Q = T^3 + Y$, the induced electric charge lattice is conventionally $\Lambda_Q \subseteq \frac{1}{3}\mathbb{Z}$ when $Q(e) = -1$ is used as the normalization convention.

2 Scope and non-claims

The theorem is conditional and fixed-domain. It classifies charge-label assignments inside a declared admissible Standard-Model bookkeeping skeleton. It does not range over arbitrary richer theories unless those theories preserve the same charge witnesses, skeleton roles, transport structure, and envelope ledger.

The theorem classifies admissible charge realizations relative to a fixed Standard-Model bookkeeping skeleton and fixed witness domain. It is not a theorem that every logically possible gauge theory reduces to the Standard Model, nor a theorem that richer ultraviolet domains cannot contain additional charge structure.

Definition 2.1 (Fixed charge scope). *A fixed charge scope is a tuple*

$$\mathcal{S}_{\text{chg}} = (\mathcal{K}, \mathcal{R}, \mathcal{O}, \mathcal{T}, \mathcal{W}_{\text{chg}}, \text{Env}),$$

where:

1. $\mathcal{K}$ is the fixed bookkeeping skeleton: field roles, chirality roles, non-abelian representation slots, interaction slots, and generation typing already licensed by upstream results.
2. $\mathcal{R}$ is the set of admissible same-scope redescription.
3. $\mathcal{O}$ is the declared composition structure for charge labels: conjugation, product neutrality, charge-sum comparison, and closed-loop transport.
4. $\mathcal{T}$ is the licensed redescription-transport family acting on skeleton roles and label presentations.
5. $\mathcal{W}_{\text{chg}}$ is the charge-witness family: neutrality of declared operators, closed-transport triviality, anomaly/tensor compatibility when the chiral gauge skeleton is declared, and boundary normalization records.
6. Env records physical inputs not derived here: the gauge skeleton, chirality content, number of Higgs doublets, whether extra $U(1)$ factors are admitted, numerical coupling values, and external matching conventions.

Definition 2.2 (Same-scope charge competitor). *A proposal P is a same-scope competitor only if it preserves $\mathcal{S}_{\text{chg}}$ up to admissible redescription. In particular, it must preserve the skeleton roles, declared charge witnesses, interaction slots, anomaly/tensor compatibility conditions, transport class, normalization ledger, no-selector discipline, and no-repair history. A proposal that adds a new $U(1)$ factor, new fermion sector, new Higgs multiplet, altered chirality skeleton, altered anomaly ledger, or new detector/witness protocol changes scope unless it factors back through the same charge quotient.*

Non-claims

This manuscript does not claim:

- a derivation of the Standard-Model gauge group from no premises;
- a derivation of numerical gauge-coupling values;
- a proof that larger-domain BSM or UV completions are impossible;
- a denial that extra $U(1)$ factors, sterile sectors, or additional fields can be studied in enlarged scopes;
- an ontic interpretation of charge labels as primitive substances;
- that quark charges are inadmissible because they are fractional in electron units;

- that additive charge offsets are allowed residual freedom;
- that empirical anomaly cancellation is imported as an external observation.

The result is narrower and stronger within its scope: in the fixed charge-assignment problem, the standing-bearing charge carrier is a rigid discrete lattice. Fractional-looking Standard-Model charges survive precisely because they are points of that lattice after primitive-unit normalization.

3 Orthodox charge interface

This section states the standard physics interface before introducing the AASC closure layer. The point is not to replace gauge theory. The point is to identify the physics object that the structural theorem classifies.

In the electroweak skeleton use the convention

$$Q = T^3 + Y, \quad (1)$$

where T^3 is the third weak-isospin generator and Y is hypercharge. For one generation, write the usual chiral fields as

field	hypercharge
Q_L	Y_Q
u_R	Y_u
d_R	Y_d
L_L	Y_L
e_R	Y_e

and let the Higgs doublet have hypercharge Y_H .

The declared charged Yukawa operators are

$$\mathcal{L}_Y \supset -\bar{Q}_L Y_u^{\text{mat}} \tilde{H} u_R - \bar{Q}_L Y_d^{\text{mat}} H d_R - \bar{L}_L Y_e^{\text{mat}} H e_R + \text{h.c.}, \quad (2)$$

where $Y_u^{\text{mat}}, Y_d^{\text{mat}}, Y_e^{\text{mat}}$ are Yukawa matrices and should not be confused with hypercharge labels. Gauge invariance under $U(1)_Y$ gives

$$Y_u = Y_Q + Y_H, \quad Y_d = Y_Q - Y_H, \quad Y_e = Y_L - Y_H. \quad (3)$$

The local anomaly-compatibility conditions for the declared chiral skeleton are

$$\begin{aligned} [SU(3)]^2 U(1) : \quad & 2Y_Q - Y_u - Y_d = 0, \\ [SU(2)]^2 U(1) : \quad & 3Y_Q + Y_L = 0, \\ [\text{grav}]^2 U(1) : \quad & 6Y_Q + 2Y_L - 3Y_u - 3Y_d - Y_e = 0, \\ [U(1)]^3 : \quad & 6Y_Q^3 + 2Y_L^3 - 3Y_u^3 - 3Y_d^3 - Y_e^3 = 0. \end{aligned} \quad (4)$$

The signs for right-handed fields reflect the standard left-handed-Weyl anomaly convention.

In the present paper these equations are not empirical inputs. They are tensor-compatibility witnesses inside a declared chiral gauge bookkeeping skeleton: the closed transport of gauge current

and phase data must not create a standing-distinguishable obstruction. If a larger theory changes the anomaly ledger by adding fields or factors, it has changed the scope.

The anomaly/Yukawa system is therefore not introduced here as an independent empirical premise selecting the lattice. It functions only as the declared tensor-compatibility witness inside the fixed Standard-Model bookkeeping skeleton. The hypercharge/Yukawa/anomaly relations do not independently generate the rigidity theorem. They instantiate the tensor-level realization witnesses for the admissible reconstruction class already fixed by the standing-preserving bookkeeping skeleton. The rigidity theorem is therefore classificatory rather than phenomenological: once that skeleton and witness burden are fixed, the Standard-Model tensor relations instantiate the unique admissible charge-lattice class up to normalization, sign, and admissible redescription rather than replacing the admissibility closure layer.

4 Charge-assignment object

Definition 4.1 (Charge-assignment object). *A charge-assignment object over $\mathcal{S}_{\text{chg}}$ is*

$$\mathcal{C} = (\mathcal{K}, \Lambda, q, \mathcal{T}, \sim_{\text{Ad}}, \mathcal{W}_{\text{chg}}, \text{Env}),$$

where Λ is the candidate charge-label carrier, $q : \mathcal{K} \rightarrow \Lambda$ assigns charge labels to skeleton roles, $\mathcal{T}$ acts by admissible redescription transport, $\sim_{\text{Ad}}$ identifies presentations with the same standing-bearing charge content, and $\mathcal{W}_{\text{chg}}$ is the declared witness family.

A charge label is not merely a number assigned to a particle name. It is a role-typed item in an admissible bookkeeping object. It must participate correctly in the interaction ledger, transport ledger, quotient ledger, and normalization ledger.

Definition 4.2 (Standing-bearing charge label). *A charge label $q(s)$ has standing when it satisfies all of the following:*

1. *it is fixed by the declared skeleton role $s \in \mathcal{K}$;*
2. *it is stable under admissible redescription of the same role;*
3. *it composes with other labels according to the declared charge operations;*
4. *it descends to the charge quotient determined by $\sim_{\text{Ad}}$;*
5. *it preserves every witness in $\mathcal{W}_{\text{chg}}$.*

Definition 4.3 (Charge witness). *A charge witness is a same-scope invariant that distinguishes or refixes charge-label content. In the Standard-Model charge scope the relevant witnesses include:*

- *operator neutrality, such as the Yukawa relations (3);*
- *conjugation compatibility, $q(\bar{s}) = -q(s)$;*
- *charge composition in products, $q(s_1 s_2) = q(s_1) + q(s_2)$ where the product is declared;*
- *closed-loop transport triviality;*

- *anomaly/tensor compatibility equations (4) when the chiral gauge skeleton is declared;*
- *boundary normalization convention, such as $Q(e) = -1$, when explicitly fixed.*

Remark 4.4 (No hidden numerical ontology). *The label carrier Λ is not introduced as an ontic substance. It is the additive carrier required for stable charge bookkeeping under the declared witnesses. Numerical presentation of labels is skin unless it descends to the standing-bearing quotient.*

5 Witness completion for charge scope

A possible escape says that a hidden charge witness may select a different charge assignment without changing the visible charge ledger. The following theorem fixes the witness-complete comparison class.

Definition 5.1 (Witness-complete charge scope). *A charge scope is witness-complete when every admissible same-scope charge distinction factors through the declared charge witnesses: operator neutrality, conjugation, closed transport, anomaly/tensor compatibility, and explicit boundary normalization. This is a comparison-class claim, not a metaphysical claim about all possible physics. Witness completeness is not an empirical completeness claim about all future physics. It is the comparison-class condition required for same-scope reconstruction classification.*

Theorem 5.2 (Charge witness completion). *Within a witness-complete charge scope, every proposed charge witness W' falls into exactly one of four classes:*

1. *W' is generated by $\mathcal{W}_{\text{chg}}$ and the charge-lattice quotient;*
2. *W' changes operator neutrality, anomaly/tensor compatibility, transport, or boundary normalization, hence changes the witness package;*
3. *W' is disconnected from same-scope charge reconstruction and has no standing force;*
4. *W' imports a hidden selector, hidden carrier, or larger-domain structure.*

No hidden witness supplies an inequivalent same-scope charge lattice.

Proof. A charge witness must either refix a standing-bearing charge distinction on the declared skeleton or not. If it refixes one using only the declared interaction, transport, anomaly, and normalization data, it is generated by $\mathcal{W}_{\text{chg}}$. If it refixes a distinction not generated by those data, then it changes a witness profile and therefore changes the object being compared. If it changes no charge distinction, it is inert skin. If it is treated as authoritative despite not being generated and not changing the witness profile, it is a hidden selector or carrier import. These alternatives exhaust the possible same-scope status of a proposed witness. □

Definition 5.3 (Charge-transport certificate). *Let $T \subseteq \mathcal{K}$ be a family of skeleton roles whose charge labels are threatened by a hidden selector, non-lattice interpolation, altered normalization, or exotic lattice representative. A charge-transport certificate for T is a same-scope witness assignment*

$$\text{Cert}_{\text{chg}} : T \longrightarrow \mathcal{W}_{\text{chg}}$$

satisfying:

- C1. Anchor re-fixation.** *Each certified label is fixed by an old skeleton role or declared composite operation.*
- C2. Witness availability.** *The certifying data lie in the declared charge witness family, not in an undeclared hidden sector.*
- C3. Transport covariance.** *Certification is preserved under admissible redescription transport and closed loops.*
- C4. Quotient preservation.** *The certified label descends to the same charge quotient $\Lambda/\sim_{\text{Ad}}$.*
- C5. No selector/no repair.** *The certificate does not choose a representative, interpolation point, lattice generator, or charge assignment not already fixed by the old witnesses.*

Proposition 5.4 (Certificate criterion for charge preservation). *A proposed charge refinement preserves same-scope standing only if it either factors through the existing charge witness package or supplies a charge-transport certificate. If neither condition holds, the proposal is selector import, repair, witness modification, or scope change.*

Proof. Standing-bearing charge preservation requires the old charge role to be re-fixed under the declared witnesses and transport relations. If the re-fixation is generated by $\mathcal{W}_{\text{chg}}$, it factors through the old witness package. If it is not generated but is nevertheless same-scope, the missing data must be supplied by a certificate satisfying anchor re-fixation, witness availability, transport covariance, quotient preservation, and no-selector/no-repair discipline. If no such certificate exists, the proposed refinement either changes what counts as a witness, selects an unwitnessed representative, repairs an undefined old assignment, or enlarges the scope. These are exactly the inadmissible or larger-domain cases. $\square$

Lemma 5.5 (No third charge status). *Inside a fixed witness-complete charge scope, any proposed charge-label candidate has exactly one admissibility status: quotient-level lattice content, normalization/sign skin, inadmissible selector/repair/non-lattice interpolation, or larger-scope structure. There is no fifth status such as fractional-but-standing outside the lattice, hidden-but-authoritative, or null-but-charge-bearing.*

Proof. If the candidate is fixed by the declared witnesses and transport relations, it descends to quotient-level charge content. If it changes no witness and no quotient-level distinction, it is skin or normalization/sign convention. If it is treated as standing-bearing without witness support, it is selector import, repair, or non-lattice interpolation. If it changes the witnesses, skeleton, abelian factors, field content, or envelope, it is larger-scope. Witness completion and bivalence of evaluated same-scope status leave no fifth admissibility-bearing role. $\square$

6 Charge-lattice normal form

The term “lattice” is used here in a precise normal-form sense.

Definition 6.1 (Charge lattice). *A charge lattice is a finitely generated additive discrete $\mathbb{Z}$ -module of admissible charge labels modulo admissible normalization/sign equivalence. In the fixed one- $U(1)$ charge scope treated here, the standing-bearing quotient is rank one: it is generated by a primitive*

unit u , is closed under charge composition and conjugation, and every standing-bearing charge label lies in

$$\Lambda = u\mathbb{Z}$$

after the chosen normalization. No metric, order, Euclidean lattice geometry, or representation-ontological structure is assumed by the word “lattice.” More generally, before choosing a primitive presentation, the charge carrier is admissibly equivalent to a rank-one discrete additive $\mathbb{Z}$ -module. In electron-normalized Standard-Model electric-charge coordinates, the minimal electric lattice is

$$\Lambda_Q \subseteq \frac{1}{3}\mathbb{Z}.$$

Remark 6.2 (Meaning of fractional). *A label may be fractional in a coarse unit and still be a lattice point. Quark charges $2/3$ and $-1/3$ in electron units are not non-lattice charges. They are $2u$ and $-u$ in the primitive unit $u = 1/3$. What is excluded is arbitrary fractional interpolation not fixed by the primitive lattice and the charge witnesses.*

Proposition 6.3 (Additive closure). *Any standing-bearing charge carrier in $\mathcal{S}_{\text{chg}}$ must be closed under addition and inverse.*

Proof. The declared charge witnesses include conjugation and neutral product formation. Conjugation requires $q(\bar{s}) = -q(s)$. Product formation requires the charge of a product to be the sum of component charges, and neutrality requires this sum to vanish for admitted scalar operators. If the carrier is not closed under inverse and addition, some admitted product or conjugate cannot be evaluated as a charge-bearing object. That is not a weaker charge assignment; it is failure of the declared charge witness. $\square$

Proposition 6.4 (Discreteness after primitive normalization). *After quotienting out global normalization, a standing-bearing same-scope charge assignment has discrete relative labels. A continuous same-scope family of relative charge labels is either skin, scope change, or inadmissible hidden structure.*

Proof. Let q_t be a continuous same-scope family of relative charge assignments preserving the skeleton and witness package. If every witness is invariant along t , then no standing-bearing charge distinction changes; by witness completion, t is skin or normalization convention. If some charge witness changes, the family does not remain same-scope. If the witness package is claimed unchanged while a standing-bearing relative charge changes, a hidden selector has been imported to distinguish labels not fixed by the witnesses. Thus no continuous family of relative charge labels survives as a second same-scope charge assignment. Once a primitive unit is fixed, the remaining standing-bearing labels lie in a discrete additive lattice. $\square$

Proposition 6.5 (No incommensurable multi-generator charge lattice in the fixed SM charge scope). *Within the fixed one- $U(1)$ Standard-Model charge scope, an incommensurable multi-generator additive carrier is not a same-scope charge lattice. It either reduces to the rank-one witness lattice, adds an extra abelian factor, or imports an unwitnessed selector.*

Proof. A rank- $r > 1$ additive carrier with incommensurable generators contains charge distinctions not expressible by a single primitive unit. If those distinctions affect product neutrality, anomaly/tensor compatibility, or transport, the witness package has changed; in gauge language this is an additional abelian charge direction or altered tensor structure. If they affect no witness,

they are unwitnessed labels and collapse to skin. If a rule selects one generator as authoritative without witness support, it is a hidden selector. Thus the fixed one- $U(1)$ scope admits only the rank-one charge lattice after quotienting normalization. $\square$

7 Transport equivariance and closed-loop triviality

The transport condition gives the lattice its standing role.

Definition 7.1 (Transport equivariance). *Let $t \in \mathcal{T}$ be an admissible redescription transport acting on a skeleton role s . A charge assignment q is transport-equivariant when*

$$q(t \cdot s) \sim_{\text{Ad}} t_{\Lambda}(q(s)) \quad (5)$$

for the induced label transport t_{Λ} .

Definition 7.2 (Closed-loop triviality). *For every closed admissible transport loop $t : s \rightarrow s$, charge labels must satisfy*

$$t_{\Lambda}(q(s)) \sim_{\text{Ad}} q(s). \quad (6)$$

A loop that produces a standing-distinguishable charge difference is standing holonomy and is inadmissible unless that holonomy is itself a declared tensor witness in an enlarged scope.

Theorem 7.3 (Transport closure for charge labels). *A same-scope charge assignment is admissible only if it is transport-equivariant and closed-loop trivial. Any charge distinction not preserved by (5) and (6) is either inadmissible, a scope change, or skin.*

Proof. A standing-bearing label must be reusable as the same charge content under admissible redescription. Equation (5) states this reuse condition for open transport; equation (6) states it for loops. If a closed loop changes the label in a way that a charge witness can detect, standing is not preserved. If the loop changes no witness, the difference is skin. If a new witness is introduced to make the difference standing-bearing, the charge scope has changed. No other same-scope status remains by witness completion. $\square$

8 No continuous or non-lattice charge deformation

Lemma 8.1 (No continuous same-scope charge orbit). *Let Λ be a standing-bearing charge carrier inside a fixed admissible bookkeeping skeleton. If a continuous deformation orbit*

$$\lambda_t \in \Lambda, \quad t \in I \subset \mathbb{R},$$

preserves all admissibility witnesses, transport relations, quotient identities, and normalization records, then the orbit is skin-level normalization or notation and contributes no tensor-level charge distinction. If the orbit changes any tensor-level witness, then it changes the reconstruction scope. Therefore no nontrivial continuous same-scope charge orbit exists.

Proof. A standing-bearing charge difference must be detected by the declared charge witnesses: operator neutrality, conjugation, closed transport, anomaly/tensor compatibility, or boundary normalization. If every such witness and transport relation is constant along λ_t , witness completion

identifies the orbit as bookkeeping skin or normalization convention. If some witness changes, the orbit no longer preserves the same charge reconstruction object. If the orbit is nevertheless treated as standing-bearing while all witnesses are held fixed, the construction has imported a selector not licensed by the scope. Thus a continuous orbit is either skin, scope change, or selector import; it is never a second admissible same-scope charge lattice. $\square$

Theorem 8.2 (Continuous charge deformation exhaustion). *Let Λ_t be a continuous family of admissible charge presentations on a fixed Standard-Model bookkeeping skeleton preserving the declared transport relations, quotient identities, normalization ledger, and charge-witness family. Then exactly one of the following holds:*

1. *the deformation is quotient-trivial and reduces to normalization/sign skin or witness-inert bookkeeping;*
2. *the deformation changes tensor-level witness structure and therefore changes reconstruction scope;*
3. *the deformation introduces undeclared selector structure and is inadmissible.*

No fourth same-scope continuous deformation class exists.

Proof. Let q_t be a continuous family with $q_0 = q$. First quotient out global normalization/sign covariance: the simultaneous transformation $q \mapsto \lambda q$ accompanied by compensating abelian coupling rescaling, together with overall sign orientation, is skin or boundary convention rather than tensor-level charge motion. After this quotient, either some relative charge coordinate changes a product-neutrality, conjugation, anomaly/tensor, closed-transport, or boundary-normalization witness, or it does not. If a witness changes, the presentation no longer preserves the same charge reconstruction object and is a scope change. If no witness changes, then the continuous variation is not standing-bearing in the charge quotient; it is bookkeeping unless a further rule declares one point of the witness-inert orbit to be authoritative. Such a rule is an undeclared selector. Thus every continuous deformation has one of the three displayed images. Fixed-domain exhaustion leaves no fourth governance-bearing locus. $\square$

Corollary 8.3 (No continuous relative charge deformation). *Within a fixed witness-complete charge scope, there is no admissible nontrivial continuous deformation of normalized relative charge labels.*

Proof. By the continuous charge deformation exhaustion theorem, a continuous deformation is quotient-trivial, witness-changing, or selector-dependent. The first case is not a new charge assignment; the second changes scope; the third is inadmissible. Hence no nontrivial same-scope continuous relative charge deformation remains. $\square$

Theorem 8.4 (No non-lattice fractional interpolation). *Fractional-looking labels are admissible exactly when they are lattice points after primitive-unit normalization. Non-lattice fractional interpolation is inadmissible in the fixed charge scope.*

Proof. Let u be the primitive charge unit. A label a/b in a coarser unit is admissible when $a/b = nu$ for some $n \in \mathbb{Z}$ after normalization. That is merely a coordinate on the lattice. By contrast, a non-lattice interpolant $q' = q + \epsilon u$ with $\epsilon \notin \mathbb{Z}$ is not generated by the primitive lattice. If q' changes a charge witness, the scope changes or the assignment fails. If q' changes no witness, the interpolant

is an unwitnessed hidden label. If a rule selects it as standing-bearing, the rule is a selector not fixed by the charge scope. Hence non-lattice fractional interpolation is inadmissible. $\square$

Corollary 8.5 (Quark fractional charges are admissible lattice points). *In the Standard-Model electric-charge presentation normalized by $Q(e) = -1$, the labels*

$$Q(u) = \frac{2}{3}, \quad Q(d) = -\frac{1}{3}, \quad Q(e) = -1, \quad Q(\nu) = 0$$

lie in $\frac{1}{3}\mathbb{Z}$. They are lattice points, not forbidden fractional interpolants.

9 Physics-facing realization: hypercharge collapse

This section supplies the concrete Standard-Model witness that the abstract charge-lattice theorem is not empty. It is conditional on the declared load: minimal chiral Standard-Model representation skeleton, one Higgs doublet, charged Yukawa operator content, and no additional compensating fields or $U(1)$ factors.

Lemma 9.1 (Yukawa compatibility). *In the convention $Q = T^3 + Y$, gauge invariance of the charged Yukawa operators gives*

$$Y_u = Y_Q + Y_H, \quad Y_d = Y_Q - Y_H, \quad Y_e = Y_L - Y_H.$$

Proof. The operator $\bar{Q}_L \tilde{H} u_R$ is neutral only if $-Y_Q - Y_H + Y_u = 0$, since $\tilde{H}$ has hypercharge $-Y_H$. The operator $\bar{Q}_L H d_R$ is neutral only if $-Y_Q + Y_H + Y_d = 0$. The operator $\bar{L}_L H e_R$ is neutral only if $-Y_L + Y_H + Y_e = 0$. Rearranging gives the displayed relations. $\square$

Lemma 9.2 (Linear anomaly collapse). *Assume the Yukawa relations and the linear anomaly equations in (4). Then*

$$Y_L = -3Y_Q, \quad Y_H = 3Y_Q.$$

Proof. The $[SU(2)]^2 U(1)$ condition gives $Y_L = -3Y_Q$. Substituting the Yukawa relations into the mixed gravitational condition gives

$$\begin{aligned} 0 &= 6Y_Q + 2Y_L - 3(Y_Q + Y_H) - 3(Y_Q - Y_H) - (Y_L - Y_H) \\ &= Y_L + Y_H. \end{aligned}$$

Thus $Y_H = -Y_L = 3Y_Q$. $\square$

Theorem 9.3 (Discrete-skeleton hypercharge collapse). *Within the declared one-generation chiral skeleton with one Higgs doublet and the charged Yukawa operators, anomaly/tensor compatibility and Yukawa compatibility force*

$$Y_L = -3Y_Q, \quad Y_e = -6Y_Q, \quad Y_H = 3Y_Q, \quad Y_u = 4Y_Q, \quad Y_d = -2Y_Q. \quad (7)$$

The family is unique up to overall hypercharge normalization and global sign convention.

Proof. From the $[SU(2)]^2U(1)$ condition, $Y_L = -3Y_Q$. From the $[SU(3)]^2U(1)$ condition, $Y_u + Y_d = 2Y_Q$. Substituting these into the mixed gravitational condition gives

$$0 = 6Y_Q + 2(-3Y_Q) - 3(Y_u + Y_d) - Y_e = -6Y_Q - Y_e,$$

so $Y_e = -6Y_Q$.

Write $Y_d = 2Y_Q - Y_u$ and substitute $Y_L = -3Y_Q$, $Y_e = -6Y_Q$ into the cubic equation. The result reduces to

$$Y_u^2 - 2Y_QY_u - 8Y_Q^2 = 0,$$

hence

$$Y_u \in \{4Y_Q, -2Y_Q\}, \quad Y_d \in \{-2Y_Q, 4Y_Q\}.$$

Thus the anomaly equations fix the unordered pair $\{Y_u, Y_d\} = \{4Y_Q, -2Y_Q\}$.

The charged-lepton Yukawa relation $Y_e = Y_L - Y_H$ gives

$$Y_H = Y_L - Y_e = (-3Y_Q) - (-6Y_Q) = 3Y_Q.$$

Then the quark Yukawa relations select

$$Y_u = Y_Q + Y_H = 4Y_Q, \quad Y_d = Y_Q - Y_H = -2Y_Q.$$

This proves (7). Multiplying all hypercharges by a common nonzero factor is normalization of the abelian generator, and reversing the abelian orientation is a global sign convention. No further relative charge freedom remains. $\square$

Corollary 9.4 (Electric charge lattice). *Choose the representative $Y_Q = 1/6$. Then*

$$Y_L = -\frac{1}{2}, \quad Y_e = -1, \quad Y_H = \frac{1}{2}, \quad Y_u = \frac{2}{3}, \quad Y_d = -\frac{1}{3}.$$

With $Q = T^3 + Y$, the electric charges of one generation are

$$Q(u) = \frac{2}{3}, \quad Q(d) = -\frac{1}{3}, \quad Q(\nu) = 0, \quad Q(e) = -1,$$

so the electric charge labels lie in the primitive lattice $\frac{1}{3}\mathbb{Z}$ in electron-normalized coordinates.

Remark 9.5 (Neutrino mass and singlets). *A gauge-singlet right-handed neutrino with $Y = 0$ is anomaly-inert in this charge theorem, and the effective Weinberg operator does not alter the charged chiral hypercharge collapse. A nonzero hypercharged new field or new abelian factor changes the envelope and is evaluated in an enlarged scope.*

10 Normalization, sign, and the rejection of additive offsets

The surviving residual freedom is not additive offset. It is global normalization and sign convention.

Theorem 10.1 (Hypercharge normalization collapse). *In the unbroken $U(1)_Y$ gauge skin, the simultaneous transformation*

$$Y \mapsto \lambda Y, \quad g_1 \mapsto g_1/\lambda, \quad \lambda \neq 0, \tag{8}$$

leaves the product g_1Y invariant in the covariant derivative. Therefore the raw normalization of Y is not a standing-bearing charge degree. Once a primitive charge unit is fixed, only global sign convention remains.

Proof. The abelian gauge interaction depends on the generator through $g_1 Y$. Transformation (8) preserves that product and therefore preserves gauge-phase coupling data in the unbroken gauge presentation. In the broken presentation, choosing a primitive unit for the resulting electric charge lattice fixes the redundancy. The remaining sign is orientation of the abelian generator or charge-conjugation convention, not a new relative charge assignment. $\square$

Normalization freedom acts only by global multiplicative rescaling of the primitive unit and by overall sign/orientation. It does not alter relative lattice position inside the admissible quotient class, and it does not generate an additive displacement of the lattice origin.

Theorem 10.2 (No additive charge offset). *An additive offset $q \mapsto q + c$ is not residual normalization freedom in the fixed Standard-Model charge scope. Unless $c = 0$, it destroys charge neutrality or changes tensor witnesses.*

Proof. Apply a common additive shift to the hypercharge relations. The relation $Y_u = Y_Q + Y_H$ would become

$$Y_u + c = (Y_Q + c) + (Y_H + c),$$

which implies the original relation only if $c = 0$. The same conclusion follows from $Y_d = Y_Q - Y_H$ and $Y_e = Y_L - Y_H$. Likewise, the neutral Higgs component condition $Q(H^0) = 0$ is not preserved by arbitrary additive shift. Additive offset is therefore not normalization; it changes the tensor-neutrality ledger. If a proposal compensates by altering operator content or representation data, it changes scope. $\square$

Corollary 10.3 (Correct residual status). *The residual charge freedom is:*

- *global normalization before primitive-unit convention is fixed;*
- *global sign or abelian-generator orientation;*
- *boundary convention, such as choosing the electron charge as -1 .*

There is no admissible additive normalization offset.

11 The charge reconstruction category

The uniqueness claim is not raw metaphysical uniqueness. It is uniqueness up to witness-preserving admissible equivalence. This section makes the comparison class explicit.

Definition 11.1 (Finite charge reconstruction object). *Let $\mathbf{C}_{\text{CHG}}$ be the category whose objects are finite charge reconstruction systems*

$$C = (\mathcal{S}_{\text{chg}}, \Lambda, q, \mathcal{O}, \mathcal{T}, \mathcal{W}_{\text{chg}}, N, \text{Env}),$$

where Λ is the additive label carrier, q is the charge assignment, $\mathcal{O}$ is the declared composition law, $\mathcal{T}$ is redescription transport, $\mathcal{W}_{\text{chg}}$ is the witness family, N records normalization/sign convention, and Env records imported physical envelope data.

Definition 11.2 (Admissible charge reconstruction morphism). *A morphism $\Phi : C \rightarrow C'$ in $\mathbf{C}_{\text{CHG}}$ is an admissible charge reconstruction map satisfying:*

- M1. Skeleton preservation.** It preserves field-role typing, interaction slots, chirality roles, non-abelian representation slots, and generation typing.
- M2. Composition preservation.** It preserves addition, inverse, conjugation, and neutrality of declared products.
- M3. Witness preservation.** It preserves $\mathcal{W}_{\text{chg}}$: operator neutrality, closed-loop transport, anomaly/tensor compatibility, and boundary normalization status.
- M4. Transport preservation.** It preserves transport equivariance and closed-loop triviality.
- M5. Normalization covariance.** It may change raw normalization only by admissible normalization/sign redescription.
- M6. Skin covariance.** It may change notation, basis, chart, field names, or representative coordinates when no charge witness changes.
- M7. No-selector/no-repair discipline.** It may not introduce a hidden charge selector, post-hoc charge repair, arbitrary fractional interpolant, or unanchored lattice representative.

Definition 11.3 (Admissible charge equivalence). *Two charge reconstruction objects C, C' are admissibly equivalent, written $C \sim_{\text{Ad}} C'$, when there are admissible morphisms both ways preserving all charge witnesses, transport status, and normalization/sign ledger. Uniqueness below means uniqueness of $[C]_{\text{Ad}}$, not uniqueness of notation.*

Proposition 11.4 ($\mathbf{C}_{\text{CHG}}$ is a category). *Finite charge reconstruction objects with admissible charge reconstruction morphisms form a category.*

Proof. The identity map preserves the skeleton, charge operations, witnesses, transport, normalization status, skin covariance, and no-selector/no-repair discipline. If $\Phi : C \rightarrow C'$ and $\Psi : C' \rightarrow C''$ are admissible, their composite preserves the same data by transitivity. A hidden selector or repair cannot appear in the composite unless it appears in one of the factors, which admissibility forbids. Associativity is inherited from ordinary map composition. $\square$

Definition 11.5 (Charge witness category and reconstruction functor). *Let $\mathbf{W}_{\text{CHG}}$ be the category whose objects are charge witness packages*

$$W_C = (\Lambda / \sim_{\text{Ad}}, \mathcal{W}_{\text{chg}}, \mathcal{T}, N),$$

with morphisms given by witness-preserving redescriptions. Define

$$F_{\text{CHG}} : \mathbf{C}_{\text{CHG}} \rightarrow \mathbf{W}_{\text{CHG}}$$

by sending a charge reconstruction object to its witness package and an admissible morphism to the induced witness-preserving map.

Proposition 11.6 (Functoriality). *F_{CHG} is a functor.*

Proof. Identity morphisms induce identity maps on witness packages. Composition of admissible charge morphisms induces componentwise composition of quotient, witness, transport, and normalization maps. Thus $F_{\text{CHG}}(\Psi \circ \Phi) = F_{\text{CHG}}(\Psi) \circ F_{\text{CHG}}(\Phi)$. $\square$

Definition 11.7 (Universal admissible charge quotient). *Let*

$$\pi_{\text{CHG}} : \mathbf{C}_{\text{CHG}} \rightarrow \mathbf{Q}_{\text{CHG}}$$

be the quotient functor identifying exactly those charge reconstruction objects and morphisms related by admissible equivalence. The category $\mathbf{Q}_{\text{CHG}}$ is the universal admissible charge reconstruction quotient of the declared scope.

The reconstruction category is required because the theorem classifies admissible equivalence classes of charge realizations rather than individual coordinate presentations. The quotient structure fixes the exact identity conditions under which rival charge assignments count as the same reconstruction object. It is not representational ornamentation: it blocks illicit selector-sensitive multiplicity by identifying witness-preserving redescrptions and separating them from witness-changing or selector-dependent scope changes.

Theorem 11.8 (Universal factorization). *For any category $\mathbf{D}$ and any functor $G : \mathbf{C}_{\text{CHG}} \rightarrow \mathbf{D}$ that is constant on admissible-equivalence classes of objects and morphisms, there exists a unique functor $\bar{G} : \mathbf{Q}_{\text{CHG}} \rightarrow \mathbf{D}$ such that*

$$G = \bar{G} \circ \pi_{\text{CHG}}.$$

Consequently every same-scope admissible charge reconstruction factors through $\mathbf{Q}_{\text{CHG}}$ up to witness-preserving bookkeeping equivalence.

Proof. Define $\bar{G}([C]_{\text{Ad}}) = G(C)$ and similarly on morphism classes. This is well-defined because G is constant on admissible-equivalence classes. The factorization is immediate from the definition of π_{CHG} , and uniqueness follows because π_{CHG} is surjective onto quotient object and morphism classes. $\square$

Definition 11.9 (Fixed-witness charge fiber). *For a declared charge witness package W_C , let $\mathbf{C}_{\text{CHG}}(W_C)$ denote the full subcategory of charge reconstruction objects inducing that same witness package under F_{CHG} .*

Theorem 11.10 (Terminal charge quotient in the fixed witness fiber). *In the quotient of $\mathbf{C}_{\text{CHG}}(W_C)$ by admissible equivalence, the charge-lattice quotient class $[C_\Lambda]_{\text{Ad}}$ is terminal in the fixed admissible charge-reconstruction category and its fixed witness fiber: every admissible same-scope charge reconstruction preserving W_C factors uniquely through $[C_\Lambda]_{\text{Ad}}$ up to witness-preserving bookkeeping equivalence. The term “terminal” is used only relative to this declared category, its witness-preserving morphisms, and its admissible-equivalence relation; it is not a claim of terminality across arbitrary mathematical charge theories or unrestricted gauge-theory space.*

Proof. By definition of the fiber, every object under comparison preserves the same charge witnesses, transport status, and normalization/sign ledger. Universal factorization through $\mathbf{Q}_{\text{CHG}}$ identifies exactly the morphisms and objects differing only by admissible redescription. Hence every object in the fiber maps to the same charge-lattice quotient class. A second terminal class would have to preserve all witnesses while differing in quotient-level charge content. That is impossible by witness completion and charge-lattice rigidity: the difference is either bookkeeping, selector import, repair, or scope change. Therefore $[C_\Lambda]_{\text{Ad}}$ is terminal in the fixed-witness quotient sense. $\square$

Corollary 11.11 (No independent same-scope rederivation). *Any independent derivation, reformulation, or presentation of the same charge witnesses either factors through the terminal charge-lattice quotient, changes the witness package, or changes scope. It does not produce a second same-scope charge realization.*

12 Partial-operator obstruction and hidden charge selectors

The category fixes same-scope equivalence. A hidden charge assignment can still be attempted as a totalization of an operator not defined by the old witness package.

Definition 12.1 (Partial charge-selector operator). *Let X_{chg} be the charge reconstruction carrier: skeleton roles, declared products, boundary normalization data, and witness records. The standing charge operator is a partial map*

$$T_{\text{chg}} : X_{\text{chg}} \rightharpoonup \Lambda$$

whose domain is the witness-fixed charge carrier: skeleton roles and certified composite operations. It is undefined on hidden labels, arbitrary fractional interpolants, and unanchored lattice representatives not fixed by the declared charge witnesses.

Remark 12.2 (Undefined is not ignorance). *Undefinedness of T_{chg} on a hidden selector is not lack of knowledge of a real charge. It is non-membership in the admissible operator domain. A larger theory may add new charge witnesses; then the scope has changed.*

A hidden selector preserving every admissible charge witness contributes no quotient-level distinction and therefore collapses to bookkeeping-equivalent presentation. If it alters admissibility-bearing reconstruction, it changes scope or imports undeclared structure.

Theorem 12.3 (Charge-selector totalization obstruction). *No admissible same-scope extension totalizes T_{chg} on hidden charge selectors, non-lattice fractional interpolants, or unanchored lattice representatives while preserving the old charge witnesses. Equivalently, there is no same-scope operator*

$$T_{\text{chg}}^* : X_{\text{chg}} \rightarrow \Lambda'$$

extending T_{chg} and adding standing-bearing hidden charge distinctions unless the distinctions are already generated by $\mathcal{W}_{\text{chg}}$ or the scope changes.

Proof. A totalization T_{chg}^* would make an old undefined charge assignment defined on the old carrier while preserving old witnesses. If the new value is fixed by the old witnesses, it was not outside the domain. If it is not fixed by the old witnesses but is treated as standing-bearing, a selector has been imported. If a new witness fixes it, the scope has changed. If it changes no witness, it is bookkeeping. Therefore no same-scope totalization survives. $\square$

Definition 12.4 (Internal charge obstruction). *For a charge reconstruction object C , define $O_{\text{chg}}(C)$ as the internal obstruction to totalizing T_{chg} on the old carrier while preserving skeleton, witnesses, transport, normalization status, and no-selector/no-repair discipline. $O_{\text{chg}}(C) = 0$ only when such a totalization exists. Otherwise $O_{\text{chg}}(C) \neq 0$.*

Remark 12.5 (Status of the obstruction object). *The obstruction object introduced here is internal to the admissible reconstruction category and is not claimed to define a standard cohomological, representation-theoretic, or lattice-homology invariant. Its role is classificatory: it records the impossibility of same-scope totalization of inequivalent charge assignments under the declared admissibility constraints.*

Theorem 12.6 (Nonvanishing of hidden-selector obstruction). *For any witness-complete fixed charge scope with a hidden selector, arbitrary non-lattice fractional interpolation, additive offset, or inequivalent unanchored lattice presented as same-scope charge content,*

$$O_{\text{chg}}(C) \neq 0.$$

Proof. If $O_{\text{chg}}(C) = 0$, then an admissible totalization exists. This contradicts the charge-selector totalization obstruction. The alleged vanishing witness must either be generated by the old charge witnesses, in which case no new charge content has been added; introduce a selector or repair, which is inadmissible; or change scope. Hence the obstruction is nonzero. $\square$

13 Finite charge-selector toy witness

The obstruction can be displayed in a finite model. The toy model is not a new physics model; it is a minimal witness for the same partial-domain closure used above. It is not intended as a phenomenological model of charge dynamics. Its role is purely obstruction-theoretic: to display explicitly how hidden interpolation and selector-sensitive totalization fail inside the fixed admissible quotient.

Definition 13.1 (Finite charge toy object). *A finite charge toy object is*

$$C_0 = (R, \Lambda_0, W_0, T_0),$$

where $R = \{r_1, r_2, r_3\}$ is a finite family of skeleton roles, $\Lambda_0 = u\mathbb{Z}$ is the primitive charge lattice, W_0 is the declared witness family of product neutrality and conjugation, and $T_0 : R \rightarrow \Lambda_0$ assigns only witness-fixed labels.

Definition 13.2 (Toy hidden interpolation). *A hidden interpolation of C_0 is an attempted extension*

$$T_0^* : R \cup \{h\} \rightarrow \Lambda'$$

with $T_0^*|_R = T_0$ and $T_0^*(h) = \alpha u$ for some $\alpha \notin \mathbb{Z}$, while h is claimed to be standing-bearing without changing W_0 .

Lemma 13.3 (Toy totalization failure). *No hidden interpolation of C_0 is an admissible same-scope charge reconstruction.*

Proof. If αu is generated by W_0 , then $\alpha \in \mathbb{Z}$ and the label was already a lattice point rather than a hidden interpolation. If $\alpha \notin \mathbb{Z}$ changes a product-neutrality or conjugation witness, then W_0 has changed and the extension is not same-scope. If $\alpha \notin \mathbb{Z}$ changes no witness but is treated as standing-bearing, the new point is selected by an unwitnessed rule. That is precisely hidden-selector totalization. If the new point is not treated as standing-bearing, it is inert skin. Thus the attempted extension is either old-lattice bookkeeping, witness change, selector import, or inert notation. $\square$

Corollary 13.4 (Finite witness for the obstruction). *The finite toy object already exhibits the full charge-selector obstruction: an old partial charge assignment cannot be totalized on a hidden non-lattice point while preserving the same witness package.*

14 Main rigidity theorem

The theorem does not prohibit rational-valued charge labels. It prohibits non-lattice interpolation and continuously deformable same-scope charge assignments. Standard quark charges are admissible because they are fixed lattice points after primitive-unit normalization.

Theorem 14.1 (Charge-lattice rigidity). *Within a fixed witness-complete admissible Standard-Model charge scope, every standing-bearing charge assignment is equivalent to a discrete additive charge lattice closed under composition, conjugation, and closed-loop transport. In the declared one- $U(1)$ Standard-Model charge scope, the quotient lattice is rank one and unique up to admissible redescription, primitive-unit normalization, and global sign convention.*

No continuous relative deformation, non-lattice fractional interpolation, additive offset, hidden charge selector, or inequivalent same-scope charge lattice yields a second admissible charge assignment. Any alleged rival is bookkeeping-equivalent, inadmissible, or larger-scope.

Proof. A standing-bearing charge label must preserve the charge witnesses and the transport quotient. Additive closure and conjugation force an additive group carrier. Transport equivariance and closed-loop triviality force the labels to descend to the admissible transport quotient. Witness completion prevents hidden witnesses from selecting additional standing-bearing charge distinctions.

Continuous relative deformations fail by the no-continuous-deformation theorem. Non-lattice fractional interpolations fail by the no-non-lattice-interpolation theorem. Additive offsets fail neutrality and tensor compatibility. Incommensurable multi-generator carriers either add an abelian charge direction, change witnesses, or become unwitnessed skin. Hidden selectors fail by the totalization obstruction.

For the Standard-Model chiral skeleton, the hypercharge-collapse theorem fixes all relative hypercharges to (7). Normalization rescaling is absorbed into the abelian coupling and primitive-unit convention; sign is abelian orientation/charge-conjugation convention. Therefore the standing-bearing charge assignment factors through a unique rank-one discrete lattice in $\mathbf{Q}_{\text{CHG}}$. $\square$

Corollary 14.2 (Charge-rigidity dependency status). *Within the declared scope:*

- **Eliminated:** *continuous relative charge deformations, non-lattice fractional interpolants, additive offsets, hidden charge selectors, unanchored exotic lattices;*
- **Forced:** *a unique discrete additive charge lattice up to admissible redescription;*
- **Certified as convention:** *global normalization before primitive-unit fixing and global sign orientation;*
- **Envelope only:** *numerical gauge coupling values and larger-domain charge sectors.*

15 Rival exhaustion

Rival route	Diagnosis	Disposition
Continuous charge deformation	Changes charge witnesses or changes nothing standing-bearing	Scope change or skin
Arbitrary fractional labels	Not generated by primitive lattice or charge witnesses	Inadmissible selector/interpolation
SM quark fractional charges	Points of $\frac{1}{3}\mathbb{Z}$ after electron-unit normalization	Admissible lattice coordinates

Additive charge offset	Violates Yukawa neutrality, $Q(H^0) = 0$, or anomaly/tensor ledger	Inadmissible
Overall rescaling	$Y \mapsto \lambda Y$, $g_1 \mapsto g_1/\lambda$ preserves $g_1 Y$	Normalization convention/skin until primitive unit fixed
Overall sign	Orientation of abelian generator or charge-conjugation convention	Boundary convention
Incommensurable lattice	Adds charge direction or unwitnessed labels	Scope change or skin
Exotic same-scope lattice	Changes relative charges or does no witness work	Tensor change or bookkeeping
Hidden charge selector	Selects labels not fixed by witnesses	Inadmissible totalization
Additional $U(1)$	Adds independent abelian charge direction and witnesses	Larger scope
Extra chiral matter	Changes anomaly/tensor ledger	Larger scope
Additional Higgs multiplet	Changes Yukawa/neutrality skeleton	Larger scope
Generation-dependent charges	Add standing-distinguishable duplication or new witness burden	Blocked by non-degeneracy unless scope enlarged
BSM or UV completion	Relevant only through induced same-scope charge quotient	Same quotient or larger-domain proposal
Anomaly-only objection	Treats anomaly equations as empirical add-ons	Recast as tensor/closed-transport witnesses in declared skeleton
Representation-theory overclaim	Says lattice presupposes representation ontology	Lattice follows from charge composition, transport, and witnesses; representation data are envelope/skeleton inputs

Theorem 15.1 (Route exhaustion). *Every same-scope attempt to avoid charge-lattice rigidity has a route image in the table above. No route survives as an inequivalent admissible same-scope charge assignment.*

Proof. A rival must alter one of the following: skeleton roles, charge operations, witness family, transport, normalization/sign ledger, hidden selector status, or envelope. If none changes, the rival is bookkeeping-equivalent. If a charge witness changes, the rival is not same-scope. If a standing-bearing label is added without a witness, it is a selector. If a failed or undefined assignment is made defined on the old carrier, it is repair or totalization. If extra fields or factors are added, the envelope changes. These cases are exactly the route table. Thus no remaining same-scope rival exists. $\square$

16 Relation to Standard-Model uniqueness

This paper supplies the charge-rigidity dependency. It is not the downstream Standard-Model uniqueness theorem by itself. Its role is to close the charge-lattice degree of freedom required by that theorem.

Dependency	Role	Charge theorem use
SM-BK-1	Admissible structure arises as standing-preserving redescription bookkeeping	Supplies the bookkeeping skeleton and redescription carrier
SM-PARAM-A	Parameters must be non-redundant and standing-preserving	Eliminates witness-inert charge parameters
SM-PARAM-B	Continuous parameters are eliminable, forced discrete, or envelope inputs by transport closure	Blocks continuous relative charge deformation
SM-PARAM-C	Generation multiplicity is finite and non-degenerate or eliminable	Blocks generation-dependent charge duplication without new witness
Charge-rigidity theorem	Charge-lattice rigidity	Supplied by this paper
Gauge-coupling parameter theorem	Raw numeric gauge couplings are not forced as unique standing-bearing numbers	Prevents overclaim from lattice rigidity to numeric coupling values
Structura	Fixed-load interior closure and hypercharge algebra	Supplies physics-facing witness for the abstract lattice theorem
downstream Standard-Model uniqueness theorem	Same-scope occupant uniqueness	Uses the charge-rigidity theorem to block exotic charge lattices and continuous charge deformation

Theorem 16.1 (Downstream authorization). *The charge-rigidity theorem authorizes the downstream Standard-Model uniqueness theorem to treat charge assignments as fixed discrete skeleton data, up to normalization/sign redescription, provided the fixed charge scope and its dependencies are retained.*

Proof. The downstream uniqueness theorem requires that a same-scope rival not alter the charge skeleton while preserving the same admissibility burden. This paper proves that any alteration of the standing-bearing charge lattice is either a witness change, hidden selector, repair, additive-offset failure, non-lattice interpolation, or larger-domain proposal. Therefore the charge-lattice degree of freedom is closed for the downstream uniqueness theorem. $\square$

17 AASC dependency map

AASC role	closure source	Imported function	Charge-domain translation
Minimal construction	admissible con-	Kernel non-optional-ity: admissibility, standing, reference, irreversibility	Charge labels must be reusable reference data, not optional notation
Standing-admissibility identity closure	iden-	No second same-domain classifier	Hidden charge classifier cannot coexist with old witness package
Boundary-trace fixation	fixation	Boundary data fix roles where declared; absent traces fail closed	Primitive-unit convention fixes normalization only when boundary record is declared
AMetric boundary	boundary	No selector, metric, ranking, typicality, or canonical representative without witness	No hidden charge selector or arbitrary fractional representative
Fixed-domain exhaustion	exhaustion	Same-scope variation must alter protected loci or be skin	Charge rivals either change witnesses/scope or are book-keeping
Closure by exhaustion for same-scope operators	for same-scope operators	Totalization and repair do not create new admissible operators	Hidden charge operators cannot define old-scope charges
Anchor/tensor/skin	skin	Exhaustive role typing	Skeleton roles are anchors; charge relations and anomaly/Yukawa compatibility are tensor; notation/normalization is skin
Gauge-coupling parameter closure	parameter closure	Raw numerical couplings are not forced standing values	Lattice rigidity does not determine g_1 , g_2 , e , or running couplings
Standard-Model classification	classification	Fixed-load occupant uniqueness depends on charge rigidity	The present theorem closes exotic charge assignments

18 Objections and replies

Objection	Closure response
“Fractional charges are real, so the theorem is false.”	The theorem does not exclude fractional-looking coordinates. It excludes non-lattice fractional interpolation. Quark charges are lattice points in $\frac{1}{3}\mathbb{Z}$ after primitive-unit normalization.

“The proof secretly assumes representation theory.”	The lattice conclusion follows from additive charge composition, conjugation, transport closure, and witnesses. Representation skeleton data are declared envelope inputs, not derived ontology.
“Anomaly cancellation is empirical physics, not AASC.”	In this scope, anomaly equations function as tensor/closed-transport compatibility witnesses inside the declared chiral gauge skeleton. A different anomaly ledger is a different scope.
“Overall normalization is a free physical parameter.”	Raw hypercharge normalization is absorbed into $g_1 Y$. Numeric gauge coupling values belong to the coupling/envelope ledger, not to charge-lattice rigidity.
“Additive offsets should survive like normalization.”	Additive offsets destroy Yukawa neutrality and neutral-Higgs/electric-charge conditions unless the offset is zero. They are not normalization.
“A hidden charge label could select another lattice.”	If it changes witnesses, scope changes. If it changes no witnesses, it is bookkeeping. If it is used as standing-bearing without witnesses, it is an inadmissible selector.
“Extra $U(1)$ models are counterexamples.”	Extra abelian factors add charge directions and witnesses. They are larger-domain theories, not same-scope rivals to the one- $U(1)$ SM charge theorem.
“BSM completions may change charge assignments.”	They may, but then they must be evaluated in the enlarged scope. If they induce the old witness quotient, they reduce to the same charge lattice on this scope.
“The theorem predicts coupling values.”	It does not. It fixes relative charge lattice structure after primitive-unit normalization. Coupling values are chart/envelope parameters.
“This is semantic.”	The theorem forbids concrete reconstruction moves: non-lattice interpolation, additive offsets, hidden selector totalization, and inequivalent same-scope lattices. It is a structural constraint on admissible charge assignments.

19 Conclusion

The charge-rigidity result developed here is compact: within a fixed admissible Standard-Model charge scope, every standing-bearing charge assignment factors through a unique discrete additive lattice up to admissible redescription, primitive-unit normalization, and global sign. Continuous relative deformation, non-lattice fractional interpolation, additive offset, hidden selector totalization, and inequivalent same-scope charge lattices are exhausted as admissible alternatives.

The contribution of the manuscript is not a reparameterization of known charge assignments. It is a fixed-domain reconstruction-classification theorem: once the declared Standard-Model bookkeeping skeleton and admissibility constraints are fixed, no second same-scope charge realization survives beyond bookkeeping-equivalent redescription. The theorem is intentionally reconstruction-theoretic. Its force lies not in deriving the Standard Model from unconstrained logical possibility, but in showing that the declared Standard-Model charge skeleton collapses to a rigid admissible lattice

class up to normalization/sign equivalence.

The physics-facing realization is the hypercharge system of the one-generation electroweak skeleton. Yukawa compatibility and anomaly/tensor constraints collapse the admissible hypercharge family to one normalization parameter. That parameter is absorbed into the $U(1)_Y$ coupling until a primitive unit is fixed. Consequently, quark charges are not a problem for the theorem: they are rational lattice points in the normalized charge lattice. The excluded alternatives are non-lattice interpolation, continuous relative charge deformation, additive offset, and hidden selector import.

This establishes the charge-lattice support theorem required by the downstream Standard-Model uniqueness architecture. In the fixed admissible charge-reconstruction category and fixed witness fiber, the charge-lattice quotient is terminal in the reconstruction sense: any same-scope derivation, reformulation, or algebraic presentation preserving the declared charge witnesses factors through the same quotient or changes scope. The result therefore blocks exotic same-scope charge assignments without overclaiming gauge-group derivation, coupling-value prediction, or beyond-Standard-Model impossibility.

A Cubic anomaly algebra

This appendix records the algebra used in the hypercharge-collapse theorem. Start with

$$Y_L = -3Y_Q, \quad Y_e = -6Y_Q, \quad Y_d = 2Y_Q - Y_u.$$

Substitute into the cubic condition:

$$\begin{aligned} 0 &= 6Y_Q^3 + 2Y_L^3 - 3Y_u^3 - 3Y_d^3 - Y_e^3 \\ &= 6Y_Q^3 + 2(-3Y_Q)^3 - 3Y_u^3 - 3(2Y_Q - Y_u)^3 - (-6Y_Q)^3 \\ &= 6Y_Q^3 - 54Y_Q^3 - 3Y_u^3 - 3(8Y_Q^3 - 12Y_Q^2Y_u + 6Y_QY_u^2 - Y_u^3) + 216Y_Q^3 \\ &= 144Y_Q^3 + 36Y_Q^2Y_u - 18Y_QY_u^2 \\ &= -18Y_Q(Y_u^2 - 2Y_QY_u - 8Y_Q^2). \end{aligned}$$

For $Y_Q \neq 0$, this yields

$$Y_u^2 - 2Y_QY_u - 8Y_Q^2 = 0,$$

so

$$Y_u = 4Y_Q \quad \text{or} \quad Y_u = -2Y_Q.$$

Then $Y_d = 2Y_Q - Y_u$ gives the unordered pair $\{Y_u, Y_d\} = \{4Y_Q, -2Y_Q\}$.

B Boundary-functional charge-sector embedding

This appendix gives a compact algebraic reading of the charge theorem. It is a boundary-functional reconstruction reading only; no DHR-style superselection classification, Haag–Kastler construction, representation-theoretic sector theorem, or new AQFT model is claimed. It records only the boundary-functional version of the same fixed-scope reconstruction theorem.

Let $\mathcal{A}_{\text{int}}$ denote the formal algebra generated by the declared interaction operators, conjugation operations, and charge-neutral composite witnesses in the fixed charge scope. A charge boundary functional is a map

$$\omega_{q \rightarrow w} : \mathcal{A}_{\text{int}} \longrightarrow V$$

that records whether a charge assignment q realizes an admitted witness $w \in \mathcal{W}_{\text{chg}}$, such as Yukawa neutrality, conjugation compatibility, closed-loop transport, or anomaly/tensor compatibility.

Definition B.1 (Boundary-functional charge sector). *Two charge presentations q and q' lie in the same boundary-functional charge sector when*

$$\omega_{q \rightarrow w} = \omega_{q' \rightarrow w}$$

for every admitted charge witness $w \in \mathcal{W}_{\text{chg}}$ after normalization/sign transport. The sector is the equivalence class of presentations preserving all declared charge functionals.

Theorem B.2 (Boundary-functional sector form of charge rigidity). *The admissible charge-lattice reconstruction determines a unique boundary-functional charge sector on the declared charge scope. A hidden selector, non-lattice interpolation, additive offset, or exotic lattice representative either changes at least one boundary-functional charge witness, is bookkeeping-equivalent inside the same sector, imports an inadmissible selector, or changes physical scope.*

Proof. The boundary functionals are precisely the algebraic representatives of the charge witness family. If a proposed presentation preserves all of them, it induces the same object of $\mathbf{W}_{\text{CHG}}$ and hence the same quotient object of $\mathbf{Q}_{\text{CHG}}$. If it changes one, the witness package has changed. If it preserves all functionals while claiming an additional standing-bearing distinction, that distinction is not fixed by the boundary-functional data and is therefore a hidden selector. If the new functional is supplied by additional fields, abelian factors, detector channels, or anomaly ledgers, the scope has changed. These alternatives exhaust the boundary-functional status of an alleged charge-sector rival. $\square$

Corollary B.3 (No hidden algebraic charge sector). *A hidden algebraic charge sector preserving all declared charge functionals contributes no quotient-level charge distinction. If it is used as standing-bearing despite that preservation, it is selector import; if it changes the functionals, it is a different sector or larger scope.*

C Notation

Symbol	Meaning
$\mathcal{S}_{\text{chg}}$	Fixed charge scope
$\mathcal{K}$	Fixed bookkeeping skeleton: field roles, interaction slots, chirality, representation slots
Λ	Candidate charge-label carrier
q	Charge assignment $q : \mathcal{K} \rightarrow \Lambda$
$\mathcal{T}$	Admissible redescription-transport family
$\mathcal{W}_{\text{chg}}$	Charge witness family

$\sim_{\text{Ad}}$	Admissible equivalence relation
u	Primitive lattice unit
Λ_Q	Electric charge lattice
$Y_Q, Y_u, Y_d, Y_L, Y_e, Y_H$	Hypercharge labels in the one-generation Standard-Model skeleton
g_1	$U(1)_Y$ gauge coupling coordinate
$\mathbf{C}_{\text{CHG}}$	Category of finite charge reconstruction objects
$\mathbf{W}_{\text{CHG}}$	Charge witness category
$\mathbf{Q}_{\text{CHG}}$	Universal admissible charge quotient
T_{chg}	Partial standing charge operator
O_{chg}	Internal obstruction to hidden-selector totalization
Env	Envelope ledger: external physical inputs not derived by this theorem

D Proof dependency graph

- G1. The fixed charge scope declares skeleton roles, charge operations, witnesses, transport, and envelope data.
- G2. A charge assignment is a role-typed object, not a bare map of names to numbers.
- G3. Witness completion blocks hidden charge witnesses inside the old scope.
- G4. Additive closure and conjugation force an additive label carrier.
- G5. Primitive-unit normalization turns admissible relative labels into a discrete lattice.
- G6. Transport equivariance and closed-loop triviality force charge labels to descend to the transport quotient.
- G7. Continuous relative deformation is skin, witness change, or selector import.
- G8. Non-lattice fractional interpolation is inadmissible; fractional-looking lattice points are admissible.
- G9. Hypercharge/Yukawa/anomaly compatibility collapses the Standard-Model relative charge family to one scale and sign.
- G10. Global normalization is absorbed into g_1 ; additive offset is rejected.
- G11. The charge reconstruction category fixes admissible equivalence.
- G12. The partial-operator obstruction blocks hidden charge-selector totalization.
- G13. The main theorem follows: the charge lattice is unique in $\mathbf{Q}_{\text{CHG}}$.
- G14. Rival exhaustion proves that every same-scope escape route is bookkeeping, inadmissible, or scope change.
- G15. The present theorem supplies the charge-rigidity dependency used by the downstream Standard-Model uniqueness theorem.

E Counterexample criterion

A same-scope counterexample to charge-lattice rigidity must exhibit a charge reconstruction object C' satisfying all of the following:

1. it preserves the same skeleton $\mathcal{K}$;
2. it preserves charge composition, conjugation, and product neutrality;
3. it preserves closed-loop transport and redescription equivalence;
4. it preserves the anomaly/tensor compatibility ledger where declared;
5. it preserves boundary normalization status;
6. it changes the charge lattice in a standing-bearing way;
7. it does not import hidden selectors, additive offsets, repair, non-lattice interpolation, extra $U(1)$ factors, extra fields, or new witnesses;
8. it does not collapse to normalization/sign convention or bookkeeping equivalence.

These conditions are jointly inconsistent. Conditions 1–5 preserve the old charge quotient; condition 6 requires changing it; condition 7 blocks the only mechanisms by which such a change could be selected; condition 8 blocks the skin and convention cases. Therefore the counterexample burden cannot be met inside the declared scope.

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